

35 Vascular Plant Structure, Growth, and Development

KEY CONCEPTS

- 35.1** Plants have a hierarchical organization consisting of organs, tissues, and cells *p.* 759
- 35.2** Different meristems generate new cells for primary and secondary growth *p.* 766
- 35.3** Primary growth lengthens roots and shoots *p.* 768
- 35.4** Secondary growth increases the diameter of stems and roots in woody plants *p.* 772
- 35.5** Growth, morphogenesis, and cell differentiation produce the plant body *p.* 775

Study Tip

Make a table: To help keep track of what different plant cells do, make the following table:

Type of plant cell	What it does	How structure fits function

Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 35
- BioFlix® Animation: Tour of a Plant Cell
- Figure 35.19 Walkthrough: Secondary Growth of a Woody Stem

For Instructors to Assign (in Item Library)

- Tutorial: Visualizing Primary and Secondary Growth
- Activity: Primary and Secondary Growth

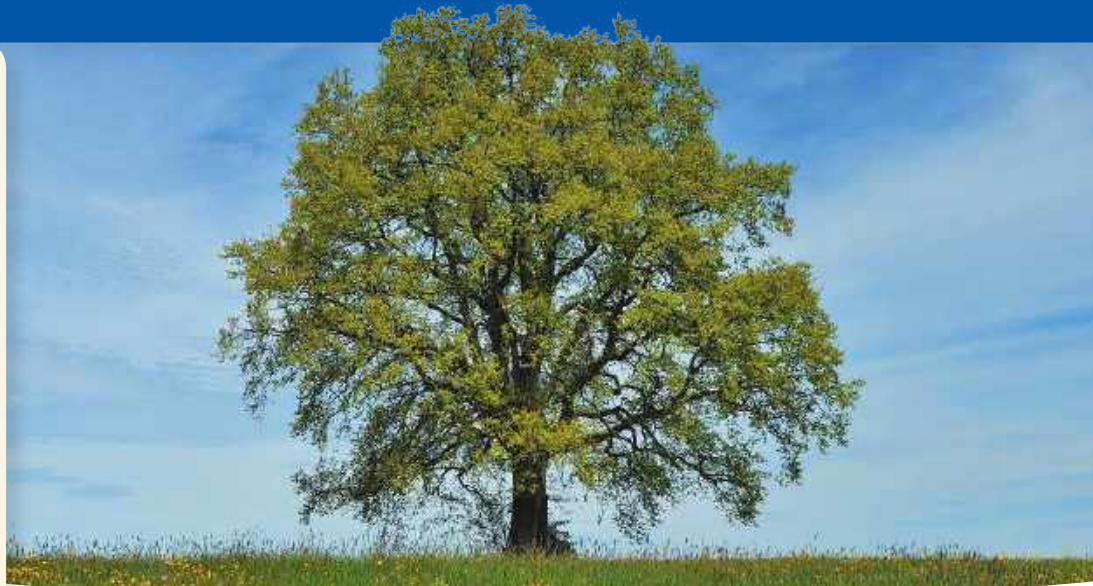


Figure 35.1 There is beauty to behold at every level of plant organization: Every cell, every tissue, and every organ has a function, and the structure of each has been molded by natural selection.

How does structure fit function in vascular plants?

At the organ level

Leaves provide surface area for absorbing sunlight and exchanging gases.

Stems

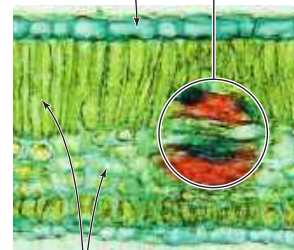
support and elevate leaves, maximizing photosynthesis.

Roots anchor the plant and absorb water and minerals.

At the tissue level

Dermal tissue protects organs.

Vascular tissue provides support and transports resources.



Leaf cross section

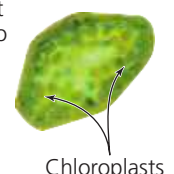
Ground tissue

includes cells that carry out photosynthesis and store sugars.

At the cellular level

Photosynthetic cells

are packed with chloroplasts that convert sunlight into chemical energy.



Chloroplasts



Tube-shaped cells transport resources. The cell shown here carries water and minerals. Others conduct sugars.

Cells with root hairs

near the tips of roots increase the surface area for absorbing water and minerals.

